LEAD GENERATION & TASK MANAGER MOBILE APP

Complete REST API Specification & Mobile Integration Architecture

TARGET PLATFORMS: iOS & Android (React Native / Flutter)

SPEC VERSION: 1.0 (Production Verified)

AUTHENTICATION: JWT Bearer Tokens (RFC 7519)

PERMISSIONS: Role-Based Access Control (RBAC) Matrix

1. AUTHENTICATION & SESSION APIS (JWT & RBAC)

METHOD	ENDPOINT & API NAME	PURPOSE & MOBILE APP USAGE	PERMISSIONS
POST	**/api/auth/token/** User Login (Obtain Tokens)	Authenticates credentials; returns JWT access & refresh tokens and user profile (ID, role, username, full name, permissions).	**Public**
POST	**/api/auth/token/refresh/** Token Refresh	Renews expired access token using valid refresh token without requiring user to re-enter username and password.	Authenticated
GET	**/api/auth/me/** User Profile / Current User	Retrieves authenticated profile, active role, assigned company memberships, and feature authorization flags.	All Roles
POST	**/api/auth/logout/** User Logout / Revoke Token	Blacklists refresh token and invalidates active session securely on server upon user signing out of mobile app.	All Roles

2. LEAD GENERATION & MANAGEMENT (CRUD, SEARCH, FILTERS) -- PART 1

METHOD	ENDPOINT & API NAME	PURPOSE & MOBILE APP USAGE	PERMISSIONS
GET	**/api/leads/** List Leads & Dynamic Counts	Paginated leads with live search (company, contact, phone, city) & multi-filters (status, segment, assignee). Returns counts.	**Admin: All** User: Assigned
GET	**/api/leads/filters/** Filter Options & Segments	Returns distinct filter dropdown options: active cities, segments with counts, categories, and assignable user lists.	All Roles
POST	**/api/leads/** Create New Lead	Creates lead; auto-preselects current user if unassigned, prevents duplicate phone/email, creates M2M assignment & task.	**Admin / Super**
GET	**/api/leads/{id}/** Lead Details & History	Returns complete lead profile with all assigned representatives, status history audit log, and assignment transition history.	Assigned / Admin

INTEGRATION NOTE: All endpoints adhere to REST conventions. Endpoints accepting JSON payloads require Content-Type: application/json and standard JWT Bearer Authorization header.

2. LEAD GENERATION & MANAGEMENT (CRUD, SEARCH, FILTERS) -- PART 2

METHOD	ENDPOINT & API NAME	PURPOSE & MOBILE APP USAGE	PERMISSIONS
PATCH	**/api/leads/{id}/** Update Lead & Assignees	Updates contact info, notes, segment. Adds or removes assigned users via assigned_to / assigned_user_ids.	**Admin: Full** User: Notes only
DELETE	**/api/leads/{id}/** Delete Lead Record	Permanently deletes lead record and safely cascades associated lead assignments and task records.	**Admin / Super**
POST	**/api/leads/{id}/status/** Quick Status Update	Updates status (Lead, HostLead, Converted, or Unassigned). Unassigning removes assignments while preserving business status.	Assigned / Admin
POST	**/api/leads/bulk-status/** Bulk Status Update	Batch updates status or unassigns multiple selected leads simultaneously with audit notes.	**Admin / Super**

3. MULTIPLE-USER LEAD ASSIGNMENT & REMOVAL

METHOD	ENDPOINT & API NAME	PURPOSE & MOBILE APP USAGE	PERMISSIONS
POST	**/api/leads/assign/** Multi-User Lead Assignment	Assigns 1+ leads to multiple users simultaneously (assigned_to_ids). Creates independent task per user.	**Admin / Super**
GET	**/api/leads/{id}/assignments/** Lead Assignees List	Returns full list of all active users assigned to this lead, including assigned_by and assigned_at timestamps.	Assigned / Admin
DELETE	**/api/leads/{id}/assignments/{uid}/** Remove User Assignment	Removes specific user from lead. Lead disappears from only that user's Task Manager; other assignees retain access.	**Admin / Super**

4. TASK MANAGER (CRUD, ASSIGNED TASKS & STATUS) -- PART 1

METHOD	ENDPOINT & API NAME	PURPOSE & MOBILE APP USAGE	PERMISSIONS
GET	**/api/tasks/** List Tasks & Status Counts	Lists tasks with status counts (All, Pending, Completed). Auto-syncs tasks for all leads assigned to authenticated user.	User: Own tasks **Admin: All tasks**
POST	**/api/tasks/** Create Task Manually	Creates task for an assigned lead with initial description, assigned date, and optional initial follow-up date.	Assigned / Admin
GET	**/api/tasks/{id}/** Task Details & History	Retrieves task details, client contact info, scheduled follow-up date, reminder date, and interaction history.	Assigned / Admin
PATCH	**/api/tasks/{id}/** Update Task Information	Edits task description, assigned date, next follow-up date, and future discussion notes.	Assigned / Admin

4. TASK MANAGER (CRUD, ASSIGNED TASKS & STATUS) -- PART 2

METHOD	ENDPOINT & API NAME	PURPOSE & MOBILE APP USAGE	PERMISSIONS
POST	**/api/tasks/{id}/status/** Toggle Task Status	Toggles or sets task status (Pending <-> Completed). Records completed_at timestamp; removes from active reminders.	Assigned / Admin
DELETE	**/api/tasks/{id}/** Delete Task	Soft-deletes task (is_deleted=True) from active manager. Preserves client profile and communication records.	Assigned / Admin
GET	**/api/tasks/available-leads/** Available Leads for Task	Returns leads assigned to the user without an active pending task, enabling one-tap creation from mobile drawer.	All Roles

5. LEAD COMMUNICATION & FOLLOW-UP (CALL, WHATSAPP, EMAIL)

METHOD	ENDPOINT & API NAME	PURPOSE & MOBILE APP USAGE	PERMISSIONS
POST	**/api/tasks/{id}/calls/** Record New Interaction	Logs Call, WhatsApp, or Email discussion notes. Schedules next follow-up date and next notes. Automatically updates task schedule and calculates reminder_date automatically.	Assigned / Admin
GET	**/api/tasks/{id}/history/** Task Communication History	Complete chronological interaction history for this user's task with interaction types, notes, dates, and agent name.	Assigned / Admin
GET	**/api/leads/{id}/interactions/** Lead 360 Interaction Log	Aggregates all interaction history across all assigned representatives for full multi-agent client visibility.	Assigned / Admin

6. REMINDERS & PUSH NOTIFICATIONS

METHOD	ENDPOINT & API NAME	PURPOSE & MOBILE APP USAGE	PERMISSIONS
GET	**/api/tasks/notifications/** Follow-up Reminders (In-App)	Returns pending tasks where reminder_date <= today (1 day prior to follow-up). Flags is_due_today & is_overdue.	User: Own tasks **Admin: Team tasks**
POST	**/api/notifications/device-token/** Register Mobile Push Token	Binds mobile FCM / APNS device token to user account for background follow-up reminder alerts and updates.	All Roles
DELETE	**/api/notifications/device-token/** Unregister Device Token	Unbinds mobile device push token upon user logout to prevent unauthorized notifications on shared mobile devices.	All Roles

AUTOMATED FOLLOW-UP REMINDER TIMELINE ALGORITHM

* Calculation Rule: reminder_date = next_followup_date - timedelta(days=1)
* Example Workflow: Call logged on 31 Oct with next follow-up on 5 Nov -> reminder_date = 4 Nov.
* Notification Alert: Representative receives in-app banner & push alert 24 hours prior to client meeting.

7. ROLE-BASED ACCESS CONTROL (RBAC) PERMISSIONS MATRIX

Standard permission enforcement across endpoints for mobile user roles

OPERATIONAL CAPABILITY	SUPER ADMIN / TECH ADMIN	ADMIN (COMPANY)	SALES EXECUTIVE
View All Leads in Organization	**YES (All)**	**YES (Company)**	NO (Assigned only)
Assign / Reassign Multiple Users	**YES**	**YES**	NO
Create New Leads Manually	**YES**	**YES**	View / Auto-Assign
Delete Leads	**YES**	**YES**	NO
Manage Own Task Manager & Status	**YES**	**YES**	**YES (Independent)**
Log Calls / WhatsApp / Email Notes	**YES**	**YES**	**YES (No overwrite)**
Independent Discussion Notes	**YES**	**YES**	**YES (Isolated)**
Receive Due / Overdue Reminders	**YES (All)**	**YES (All)**	**YES (Own tasks only)**

8. MOBILE APPLICATION INTEGRATION & ARCHITECTURE GUIDELINES

1. JWT AUTHENTICATION HEADER

* Header: Authorization: Bearer <access_token>
* Include header in every authorized API request.
* On HTTP 401 Unauthorized, call token refresh.
* If refresh fails, route user back to Login screen.
* Secure tokens in OS Keychain / EncryptedSharedPreferences.

2. MULTI-USER ASSIGNMENT MODEL

* Leads link to users via lead_assignments table.
* Multiple reps see the same lead simultaneously.
* Each rep maintains an independent LeadTask record.
* Rep follow-up notes NEVER overwrite other reps.
* Unassigning rep removes lead from ONLY that rep.

3. ACTION LINKS & DEEP LINKING

* Direct Phone: tel:<phone> launches native dialer.
* WhatsApp: wa.me/<num>?text=Hello%20<name>
* Email: mailto:<email>?subject=Follow-up...
* Mobile app can launch communication apps directly
* Automatically prompts rep to log notes afterward.

4. STATUS CODES & ERROR RESPONSES

* 200 OK / 201 Created: Standard JSON payload.
* 400 Bad Request: Validation error details.
* 401 Unauthorized: Invalid or expired token.
* 403 Forbidden: Insufficient role permissions.
* 409 Conflict: Duplicate phone or email exists.

DOCUMENT STATUS: APPROVED FOR MOBILE APPLICATION IMPLEMENTATION

BACKEND ENGINE: Django 5.x / Django REST Framework / MySQL Database

MOBILE PLATFORMS: iOS 14+ / Android 10+ (React Native / Flutter Cross-Platform)

SECURITY & COMPLIANCE: Encrypted Token Storage, Role-Based Access Control (RBAC), Audit Trails

VERIFICATION: Full compatibility with existing Impression Web CRM database and APIs.